

Lecture 2: Population Inference for Proportions

August 18, 2023

Review

- ▶ Last lecture, we discussed description/summarization with means, conditional means, and differences in conditional means.
- ▶ We considered the properties of these summaries (e.g., balance point, least squares, etc.) and how much was left out by summarizing (e.g., squared residuals)
- ▶ This was done in the context where we only cared about the sample, either provisionally or because we had sampled everyone (i.e., a census).

Today

- ▶ Today we consider the context where we have a random sample from a large population and we have measured a binary outcome.
- ▶ Our goal over the next couple of lectures is to summarize the population with population means, conditional population means, and differences in conditional population means.
- ▶ Summarizing data we don't have (everything in the population that isn't in the sample) is more difficult than summarizing data we do have (sample).

Sample

Suppose that a week before the 2020 presidential election, you contacted a sample of $n = 625$ potential Georgia voters, randomly selected (with replacement) from the population of $N = 7,233,584$ on the public voters register, to ask whether they planned to vote.

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Table: Sample

i	1	2	3	4	...	625	$\bar{y}_{625}$
y_i	1	1	0	1	...	0	.66

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i	1	2	3	4	...	625	$\bar{y}_{625}$
y_i	1	1	0	1	...	0	.66

Remember that the mean of a binary variable is a proportion.

Sample versus Population

Table: Sample

i	1	2	3	4	...	625	$\bar{y}_{625}$
y_i	1	1	0	1	...	0	.68

Sample versus Population

Table: Sample

i	1	2	3	4	...	625	$\bar{y}_{625}$
y_i	1	1	0	1	...	0	.68

After election day, we found that in fact 4, 999, 960 people, or roughly 70% of registered voters, turned out to vote.

Table: Population

j	1	2	3	4	...	...	...	...	7,233,584	$\bar{y}_{7,233,584}$
y_j	0	1	0	1	...	...	...	...	1	.70

Two Questions

1. .68 is close to .70, did we get lucky?
2. Could we have predicted how close we would get before seeing the .70?

Sampling Distribution of $\bar{Y}_{625}$

i	1		2		...	625		$\bar{Y}_{625}$
	J_1	Y_1	J_2	Y_2	...	J_{625}	Y_{625}	
	9562350	1	8763351	1	...	1294801	0	.68

Sampling Distribution of $\bar{Y}_{625}$

i	1		2		...	625		$\bar{Y}_{625}$
s	J_1	Y_1	J_2	Y_2	...	J_{625}	Y_{625}	
1	9562350	1	8763351	1	...	1294801	0	.68
2	5331704	0	4533839	1	...	3342359	1	.70

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3	5129936	0	10981600	0	...	4096184	1	.75
4	803605	0	7036389	1	...	803605	0	.73
5	148567	0	3833847	1	...	4769869	1	.69
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1 mil	4163458	0	8384613	1	...	377981	1	.74
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$f(\cdot)$								

- What is the distribution of Y_1 ?

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$f.(.)$	Bern(.70)		Bern(.70)			Bern(.70)		

- ▶ What is the distribution of Y_1 ?
- ▶ What is the distribution of $\sum_i^{625} Y_i$?

Sampling Distribution of $\bar{Y}_{625}$

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⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
$f(\cdot)$	Bern(.70)		Bern(.70)			Bern(.70)		$\frac{Bin(625, .70)}{625}$

- ▶ What is the distribution of Y_1 ?
- ▶ What is the distribution of $\sum_i^{625} Y_i$?
- ▶ What two numbers show that we sampled with replacement?

Review of Bernoulli and Binomial Distributions

We can write distributions for our random variables (RVs) that describe the probability of getting particular values for these variables.

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$Y_1, \dots, Y_{625}$ are Bernoulli RVs:

$$f_{Y_1}(y_1) = (.71)^{y_1} (.29)^{1-y_1} \text{ for } y_1 = 0, 1$$

$$\vdots = \vdots$$

$$f_{Y_{625}}(y_{625}) = (.71)^{y_{625}} (.29)^{1-y_{625}} \text{ for } y_{625} = 0, 1$$

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$Y_1, \dots, Y_{625}$ are Bernoulli RVs:

$$\begin{aligned} f_{Y_1}(y_1) &= (.71)^{y_1} (.29)^{1-y_1} \text{ for } y_1 = 0, 1 \\ \vdots &= \vdots \\ f_{Y_{625}}(y_{625}) &= (.71)^{y_{625}} (.29)^{1-y_{625}} \text{ for } y_{625} = 0, 1 \end{aligned}$$

$\sum_{i=1}^{625} Y_i$ is a Binomial RV:

$$f_{\sum_{i=1}^{625} Y_i}(\sum_{i=1}^{625} y_i) = \binom{625}{k} (.70)^{\sum_{i=1}^{625} y_i} (.30)^{625 - \sum_{i=1}^{625} y_i} \text{ for } \sum_{i=1}^{625} y_i = 0, \dots, 625$$

The Sampling Distribution in $\mathbb{R}$

```
# Resample the number of voters 1,000,000  
# times and store these 1,000,000  
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The Sampling Distribution in R

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# Divide all of these numbers  
# by the sample size.  
  
Ybar_vec <- sumY_vec/625
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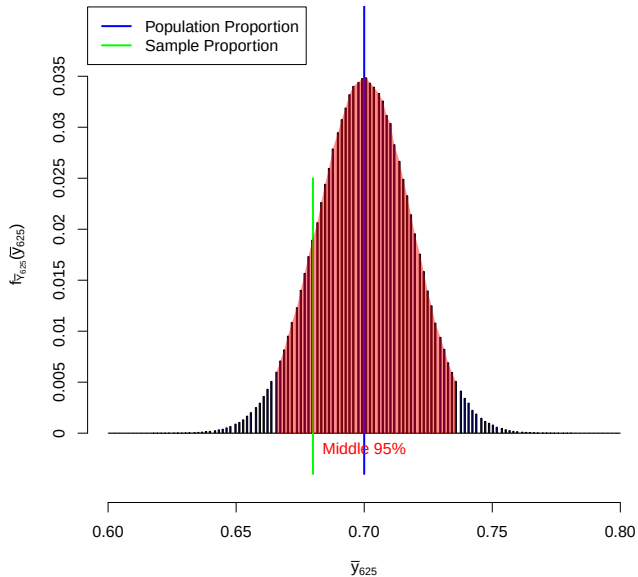
# Divide all of these numbers
# by the sample size.

Ybar_vec <- sumY_vec/625

# Plot a histogram

hist(Ybar_vec)
```

Sampling Distribution of $\bar{Y}_{625}$



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Predicting before election day

Could we have predicted how close we would get? Yes, in a sense.

- ▶ We will use a confidence interval.
- ▶ A confidence interval has a width, which answers the “how close” question.
- ▶ However, a confidence interval also has a coverage probability. This probability will almost never be 100%.

CI Coverage of $\bar{Y}_{625} \pm .01$

As an exercise, let's arbitrarily choose a width of 2% and use the sampling distribution.

i	1 Y_1	2 Y_2	...	625 Y_{625}	$\bar{Y}_{625}$	$\bar{Y}_{625} \pm .01$
	1	1	...	0	.68	[.67,.69]

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2	0	1	...	1	.70	[.69,.71]
3	0	0	...	1	.75	[.74,.76]
4	0	1	...	0	.73	[.72,.74]
5	0	1	...	1	.69	[.68,.70]
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	0	1	...	1	.74	[.73,.75]
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Coverage of $\overline{Y}_{625} \pm .01$ in R

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# Divide all of these numbers
# by the sample size.

Ybar_vec <- sumY_vec/625

# Calculate coverage

mean(Ybar_vec-.01 <= .7 & Ybar_vec+.01 >= .7 )
# about 40\%
```


Calibrating the CI

The previous exercise doesn't address our question because

1. We won't have access to the sampling distribution before election day.
2. We don't want to choose the width and receive the coverage, we want to choose coverage and receive the width

There are many ways to calibrate a CI (choose a coverage, receive the width) without observing the sampling distribution. We will explore the percentile bootstrapping approach for getting a CI with 95% coverage for the remainder of this lecture.

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However, we can take repeated random samples *with replacement* of size 625 from the sample of size 625.

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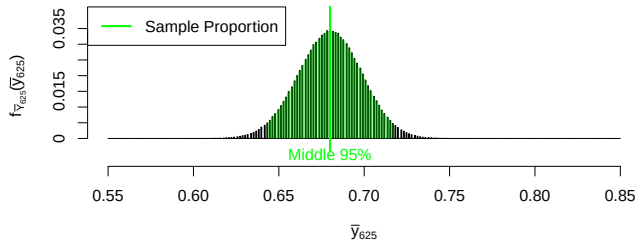
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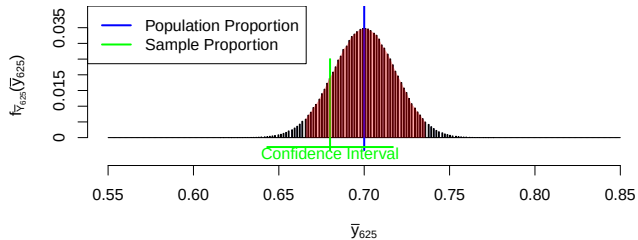
This is equivalent to replacing .70 with .68 in the R code.

```
sumY_vec <- rbinom(1000000, size=625, prob=.68)
```

Estimated Sampling Distribution of $\bar{Y}_{625}$



Sampling Distribution of $\bar{Y}_{625}$



The Logic of Confidence Intervals

If the bootstrapped sampling distribution has the same shape as the sampling distribution,

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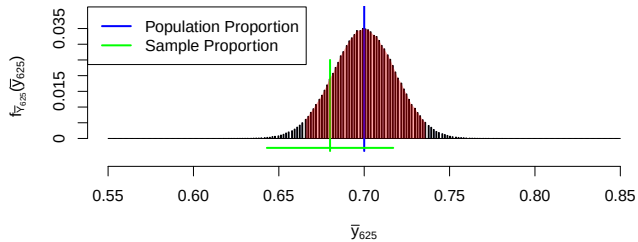
1. The 95% confidence interval will *cover* the population proportion when the sample proportion is in the middle 95% of the sampling distribution.
2. The 95% confidence interval will *not* cover the population proportion when the sample proportion is not in the middle 95% of the sampling distribution.

The Logic of Confidence Intervals

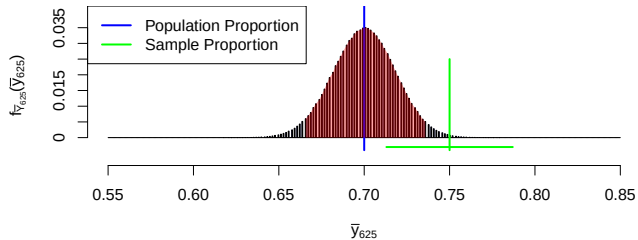
If the bootstrapped sampling distribution has the same shape as the sampling distribution,

1. The 95% confidence interval will *cover* the population proportion when the sample proportion is in the middle 95% of the sampling distribution.
2. The 95% confidence interval will *not* cover the population proportion when the sample proportion is not in the middle 95% of the sampling distribution.
3. There is a 95% probability of getting a sample proportion in the middle 95% of the sampling distribution, therefore there is 95% probability of getting a 95% confidence interval that covers the population proportion.

Sampling Distribution of $\bar{Y}_{625}$



Sampling Distribution of $\bar{Y}_{625}$



The Sampling Distribution of the Confidence Interval

i s	1 Y_1	2 Y_2	...	625 Y_{625}	$\bar{Y}_{625}$	95% CI
1	1	1	...	0	.68	
2	0	0	...	1	.75	
3	0	1	...	1	.70	
4	0	1	...	1	.73	
5	0	1	...	1	.69	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
1 mil	0	1	...	1	.74	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Table: Sampling Distributions for $\bar{Y}_{625}$ and the 95% Confidence Interval

95% of the rows in this sampling distribution will have a CI that covers .70.

The Sampling Distribution of the Confidence Interval

i	1	2	...	625	$\bar{Y}_{625}$	95% CI
s	Y_1	Y_2	...	Y_{625}		
1	1	1	...	0	.68	[.643,.717]
2	0	0	...	1	.75	
3	0	1	...	1	.70	
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$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
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3	0	1	...	1	.70	[.663,.737]
4	0	1	...	1	.73	[.694,.766]
5	0	1	...	1	.69	[.653,.727]
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	0	1	...	1	.74	[.705,.775]
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Table: *Sampling Distributions for $\bar{Y}_{625}$ and the 95% Confidence Interval*

95% of the rows in this sampling distribution will have a CI that covers .70.

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Informal logic:

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4. The variance of the sampling distribution of the proportion is $\sqrt{.7(1 - .7)/625} = .0183$.

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3. The variance fully characterizes the shape of a normal distribution (not the location).
4. The variance of the sampling distribution of the proportion is $\sqrt{.7(1 - .7)/625} = .0183$.
5. The variance of the bootstrapped sampling distribution of the proportion is $\sqrt{.68(1 - .68)/625} = .0187$.

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2. We can observe that the bootstrapped sampling distribution of the proportion is approximately normally distributed.
3. The variance fully characterizes the shape of a normal distribution (not the location).
4. The variance of the sampling distribution of the proportion is $\sqrt{.7(1 - .7)/625} = .0183$.
5. The variance of the bootstrapped sampling distribution of the proportion is $\sqrt{.68(1 - .68)/625} = .0187$.
6. When n is large, these variances are close.

Two Questions

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Summary of Population Inference for Proportions

